Yujen Lan

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BMC / OpenBMC Platform-Management Software Engineer — Redfish service development & DMTF compliance specialist. Cut BMC startup time by 60% and led a large-scale automated validation framework covering 40+ resource types across 10 platform configurations.

Summary

- Software Engineer with 3+ years of experience developing Linux-based platform management software using Modern C++, Python, and OpenBMC technologies.
- Experienced in **DMTF Redfish, OpenBMC**, API development, platform management services, and software architecture modernization. Proven track record of improving performance, maintainability, interoperability, and standards compliance across BMC platforms.
- Identified structural blind spots in an industry-standard validation tool (DMTF Redfish Validator) — it couldn't verify field presence or operation ordering, only formatting — and designed a **new validation engine** to close the gap. Now covers 41 resource types and ~690 checks across real hardware, replacing ~40 manually-maintained modules.

Technical Skills

Software Engineering: Software Architecture | API Design | OOP | Design Patterns | RAII | Concurrency & Asynchronous Programming | Performance Optimization | System-Level Debugging | Refactoring
Platform Management Ecosystem: DMTF Redfish | OpenBMC | bmcweb | MCTP | PLDM | IPMI | DBus | Yocto / BitBake
Languages & Tools: C++, C, Python, Bash | CMake, Meson, CI/CD | Linux, Systemd | Git

Experience

Insyde Software | Software Engineer *July 2024 – Present*

- **Reduced BMC startup time by over 60%** through startup profiling, dependency analysis, and service initialization optimization.
- **Designed a sustained integration workflow with upstream OpenBMC** that reduced porting overhead and patching effort from approximately months weeks.
- **Modernized legacy platform-management services** into modular Modern C++ and DBus-based architectures, improving maintainability.
- **Refactored Redfish schemas and service implementations**, improving maintainability, readability, and runtime performance by approximately 30%.
- **Improved DMTF Redfish compliance and interoperability** by resolving service-validator findings and protocol compatibility issues.
- **Built reusable utility libraries** for resource management, file operations, and lifecycle control using RAII and objectoriented design principles.
- **Resolved production stability issues** including service crashes, object lifetime defects, resource ownership problems, and concurrency-related issues.
- **Reorganized OpenBMC meta-layer** structures and multi-repository projects while preserving development history and improving maintainability.
- **Upgraded legacy back-end service** architecture to a lightweight reverse-proxy model, improving deployment flexibility and operational simplicity.
- **Simplified firmware update workflows** through unified API-driven multipart update operations, improving usability and deployment efficiency.

- **Lead development of an end-to-end Redfish validation platform** capable of validating GET, PATCH, POST, and OEM workflows beyond existing industry validation tools — covering 40+ resource types across ~690 tests and 10 platform configurations.
  GitHub: https://github.com/JohnBlue-git/RedfishAutoYamlTestInterview
- Diagnose and resolve cross-layer issues spanning Redfish services, DBus services, platform middleware, Linux systems, networking, and firmware components.
- Collaborate with software and firmware teams to deliver platform-management features and improve overall system reliability.
- Author technical API documentation and developer guides for internal teams and OEM partners.

SinoPac Securities | Software Engineer *June 2023 – April 2024*

- Developed and maintained backend applications using C++ on Linux platforms.
- **Improved data-processing efficiency by 3 to 4 times** through algorithm optimization and workflow redesign.
- Built automated CI/CD pipelines integrating build validation and testing processes, reducing manual effort and improving software quality.

Education

National Cheng Kung University (NCKU)

Master Degree *in Systems and Naval Mechatronic Engineering (Year: 2021 GPA: 4.3)*

Bachelor Degree *in Systems and Naval Mechatronic Engineering (Year: 2018 GPA: 3.7)*

Open Source Projects

MultiFish

Experimental project exploring REST API architecture, backend service design, and modern Golang system development.
Github: https://github.com/JohnBlue-git/MultiFish

HowToSDBusPlus

Practical guide and implementation examples for modern C++ DBus service development, inter-process communication, and Linux system programming.
GitHub: https://github.com/JohnBlue-git/HowToSDBusPlus

GoVarlink

Implementation examples and experiments focused on lightweight IPC mechanisms and service communication.
GitHub: https://github.com/JohnBlue-git/GoVarlink

HowToYocto

Documentation and practical examples covering Linux build systems, software integration, and platform customization.
GitHub: https://github.com/JohnBlue-git/HowToYocto

HowToPLDM

Technical notes and implementation examples focused on protocol architecture and system integration concepts.
GitHub: https://github.com/JohnBlue-git/HowToPLDM